SAMANTHA SCOTT

Computer Engineering Senior • Software Development • Typescript • Python • C/C++
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PROFESSIONAL SUMMARY

Computer Engineering senior at Florida International University (GPA: 3.99), graduating December 2026 and recognized as the Fall 2026 Outstanding Bachelor's Graduate. Hands-on software development experience in TypeScript, React Native, Python, C/C++, Supabase, and Git. Developed mobile, Linux-based, and cross-platform applications involving real-time data, database integration, and system APIs. Experienced in software debugging, testing, and collaborative development.

TECHNICAL SKILLS

Programming : Typescript, JavaScript, C, C++, Python, Java, VHDL, Assembly
Application & Data: React Native, Expo, Supabase, JSON, real-time data subscriptions, database integration, API integration
Developer Tools: Git, GitHub, GitHub Actions, VS Code, Linux, Raspberry Pi OS, systemd, SSH, UFW, Vivado/Vitis
Computer Science: Data structure and algorithms, Operating Systems, debugging, root-cause analysis
Embedded & Networking: Raspberry Pi, ESP32, Arduino, ARM Cortex-M, BLE, GPIO, UART, SPI, I2C, TCP/IP fundamentals

RELEVANT PROJECTS

DriveSense - Biometric Driver Monitoring System | *React Native, TypeScript, Supabase, Raspberry Pi* **2026 – Present**

- Develop a React Native/TypeScript mobile application integrating real-time biometric telemetry, driver profiles, emergency contacts, and automated safety workflows.
- Integrate Supabase for user data management, real-time subscriptions, incident logging, and synchronization with embedded hardware.
- Implement Sensi, an AI-assisted safety system featuring on-device language-model inference, offline speech recognition, text-to-speech, and event-triggered emergency assistance.
- Develop and test application functionality across device connectivity, user interaction, and emergency-response workflows.
- Validated application functionality using automated unit tests, TypeScript checks, and GitHub Actions CI workflows to identify regressions and maintain code quality.

Cross-Platform File Management & Locking Utility | *C++, POSIX, Windows API* **July 2026**

- Built a cross-platform C++ utility using native operating-system calls for direct-access file I/O, file positioning, and file lifecycle management.
- Implemented shared/exclusive byte-range locking and lock tracking with POSIX fcntl() and Windows LockFileEx() APIs for multi-process file-access testing.
- Tested file access across Windows and Linux, validating locking, error handling, and platform-specific system calls.

Raspberry Pi Linux Server Monitor | *Python, Linux, Raspberry Pi, systemd, SSH, UFW* **Sep 2026**

- Built and deployed a Python health-monitoring tool that collects CPU, memory, disk, temperature, and uptime data directly from Linux system interfaces.
- Implemented warning thresholds and timestamped logs; automated five-minute checks with a systemd service/timer and secured remote administration with SSH and UFW.

Traffic Light Controller with Pedestrian Interface | *Zybo Z7 FPGA, VHDL, C* **Apr 2026**

- Designed and validated a deterministic multi-state controller in VHDL with pedestrian-request logic, timing controls, and OLED feedback; synthesized, deployed, and tested it on FPGA hardware.

PROFESSIONAL EXPERIENCE

Lead Student Assistant – IT | *Florida International University | Miami, FL* **Aug 2026 – Present**

- Lead three student assistants in diagnosing technical issues, coordinating support activities, and resolving hardware, software, and network-related incidents.
- Coordinate the deployment and testing of 200 electronic access-control devices, troubleshooting configuration, firmware, and connectivity issues.

Student Assistant – IT | *Florida International University | Miami, FL* **Oct 2024 – Aug 2026**

- Supported network-connected building systems across 9 residential buildings, diagnosing controller, gateway, firmware, configuration, and connectivity issues.
- Administered and supported 200+ endpoints through Active Directory, SCCM, and JAMF; documented resolutions for hardware, software, and network faults.
- Maintained Xibo digital signage and deployed technology by troubleshooting endpoint connectivity and updating BIOS, firmware, and device drivers.

Student Assistant – Dean of Student Services | *Miami Dade College | Miami, FL* **Nov 2022 – Apr 2024**

- Coordinated records, official correspondence, and campus operations with careful attention to accuracy, deadlines, and cross-functional communication.

Customer Service Officer / Head Teller | *Bank of Nova Scotia Jamaica Ltd | Jamaica* **Feb 2016 – Aug 2022**

- Promoted to Head Teller; trained and supervised a team of four while ensuring regulatory compliance, accurate daily balancing, and transaction accuracy.

EDUCATION

Florida International University | *Miami, FL* **Expected Dec 2026**
B.S. in Computer Engineering | GPA: 3.99/4.00
Honors & Awards: Outstanding Bachelor's Graduate Award (Fall 2026), Dean's List
Relevant Coursework: Data Structures, Systems Programming, Operating Systems, C++ Programming for Embedded Systems, Python Programming in ECE, Telecommunication Networks